



Research paper

A machine learning approach to quality control oceanographic data

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ABSTRACT

Sensor errors are inevitable when measuring the ocean; thus, a reliable dataset of observations requires a quality control (QC) procedure capable of detecting spurious measurements. While manual QC by human experts minimizes errors, it is inefficient to handle large datasets and vulnerable to inconsistencies between different experts. Although automatic QC addresses some of these issues, the traditional methods result in high rates of false positives. Here, I propose a machine learning approach to automatically QC oceanographic data based on the Anomaly Detection technique. Multiple tests are combined into a single multidimensional criterion that learns the behavior of the valid measurements and identifies bad samples as outliers. When applied to 13 years of hydrographic profiles, the Anomaly Detection resulted in the best classification performance, reducing the error by at least 50%. The Anomaly Detection approach introduced here was implemented in the Python package CoTeDe, an open-source framework to quality control oceanographic data.

1. Introduction

Conservation of momentum, heat, and mass in the ocean depend on the seawater density (ρ) (see Gill, 1982); thus, ρ is an essential variable to understand and describe oceanographic process. Because variations of ρ in the ocean are usually small and accelerations are relatively large, direct *in-situ* measurements of density along the water column are impractical (Backer Jr., 1981). As an alternative, oceanographers infer ρ from temperature (T), salinity (S), and pressure (P), using a Gibbs function formulation for seawater (Feistel, 2008; Millero et al., 2008; Backer Jr., 1981). In the context of global warming, the vertical structure of $\{T, S, P\}$ is key for computing heat budgets and sea level rise associated with thermal expansion. Thus, our ability to understand, describe, and predict ocean warming depends on the quality of observations of temperature, salinity, and pressure. For example, a persistent bias on T profiles from expendable Bathy-Thermographs (XBT) resulted in errors of up to 50% in the estimates of ocean warming and thermal expansion in recent decades (Cowley et al., 2013; Domingues et al., 2008; Levitus et al., 2009), which further emphasizes our need for high-quality measurements.

The marine environment is harsh for electronic sensors, making it inevitable to have some spurious measurements. In response to that, data distribution centers provide measurements with quality flags from well-established quality control (QC) procedures. Some of the widely used procedures have been defined by: The Global Temperature and Salinity Profile Programme (GTSP, UNESCO-IOC, 2010), the European Global Ocean Observing System (EuroGOOS) of the Intergovernmental Oceanographic Commission of UNESCO (IOC-GOOS, DATA-MEQ

working group, 2010), the Argo Program (Wong et al., 2020), the CSIRO XBT Program (Bailey et al., 1994), the Australian Integrated Marine Observing System (Morello et al., 2011, 2014), and the Integrated Ocean Observing System (QARTOD group, 2016). While each type of sensor requires some specific steps in the QC procedure, the recommendations mentioned above have several tests in common and share the same general structure of a sequence of independent tests. A major weakness of this approach is the lack of context awareness to classify the data (Smith et al., 2012), often compensated by complementary manual QC. Given the adaptive skills of the human brain in pattern recognition and the relatively limited amount of hydrographic measurements available, manual evaluation by experts is currently the best option to minimize QC errors. However, the efficiency of an automatic QC better suits real-time operations, such as weather forecast and the processing of large datasets. Also, inconsistencies in manual QC may pose a problem to compare or to integrate multiple datasets, classified by different experts (Morello et al., 2014). Thus, improvements in measuring the ocean must be followed by advances in oceanographic QC methods that reduce the burden on the human experts without compromising the classification skill, and that allow efficient and coherent data aggregation.

Developments in data mining and machine learning have been revolutionizing data analysis (Ivezić et al., 2014). Such techniques provide a convenient framework to improve automatic QC by using supervised learning to reduce the discrepancy with the human expert evaluation. From a machine learning perspective, QC of oceanographic

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data is a classification task where the simplest setup is composed of only two categories: good or bad data. A powerful technique used for classification is Bayesian Networks (Smith et al., 2012), which combines different factors through an empirical network of relations to perform a quality classification by statistical inference. Such intricate decision making comes with a price: a slow learning rate that must overcome the natural variability. Therefore, the calibration of Bayesian Networks requires high sampling rates, which results in a large volume of data (Bettencourt et al., 2007). Coastal monitoring sensors that have a fixed position may satisfy this requirement, allowing for a skillful QC classification using Bayesian Network (e.g., Smith et al., 2012; Rahman et al., 2013, 2014) and improvements for similar cases could be achieved from tuning the Bayesian Network rather than employing an entirely different technique. That is not necessarily true for other types of sensors with different sampling strategies.

Because modern sampling procedures are designed to minimize spurious measurements, the QC of hydrographic data is an unbalanced classification problem. The number of good measurements is typically at least two orders of magnitude larger than the available bad samples, which critically compromises the calibration of most machine learning techniques, including Bayesian Networks. A major contribution to this problem came from Rahman et al. (2014), where the authors applied cluster undersampling to overcome the unbalancing issue. However, cluster undersampling can only fully address the unbalanced classification problem if the data used in the calibration phase statistically represents each one of the categories being classified, which is particularly challenging for oceanographic data that are sparse in space and time. Spurious data have no bounds of feasibility, being, therefore, relatively more variable than valid data, while corresponding only to a small fraction of the measurements. If the available samples of bad data do not statistically represent all possible types of spurious measurements, the calibration would minimize the classification error for that particular dataset, but it would not necessarily be a good predictor for new bad samples, an issue known as overfitting (Ivezić et al., 2014).

A machine learning technique more adequate to QC oceanographic data is the Anomaly Detection, in which the model learns the behavior of the good data and identifies the bad data as outliers. Therefore, neither the relative nor the absolute sample size of the bad data compromises the QC classification, allowing identification of spurious measurements even if that kind of error has never been observed. Previous implementations of Anomaly Detection for environmental measurements were based on comparisons between multiple synchronous sensors (Bettencourt et al., 2007) or auto-regressive models from individual sensors (Hill and Minsker, 2010), both requiring stationary statistics, which is often not the case for hydrographic profiles. Here, I propose a novel approach to search for outliers in several characteristics of the data, or features, instead of the measured value itself. This is somewhat aligned with the vision of Gronell and Wijffels (2008), but with the advantages of the Anomaly Detection framework.

Although most observing programs provide data with quality flags, observational studies led by individuals or small groups often require some quality control before the data is ready to be used. Non-operational programs can rarely afford a QC expert in their team, compromising the quality of their data products. This manuscript introduces a new method to quality control oceanographic data based on the Anomaly Detection technique, which reduces classification mistakes. The method presented here was implemented in the open-source QC framework CoTeDe (Castelão, 2020), offering a collection of pre-sets which makes it convenient to be adopted by data centers with large volumes of data, while also being flexible enough to accommodate the fine-tuning required by specific studies. Here, the performance of the Anomaly Detection is compared with state of the art automatic QC in a case-study using CoTeDe on a hydrographic dataset described in Section 2. Traditional QC procedures are briefly reviewed in Section 3.1, and the Anomaly Detection is introduced in Section 3.2. Results and discussion are presented in 4 while Section 5 offers some concluding remarks.

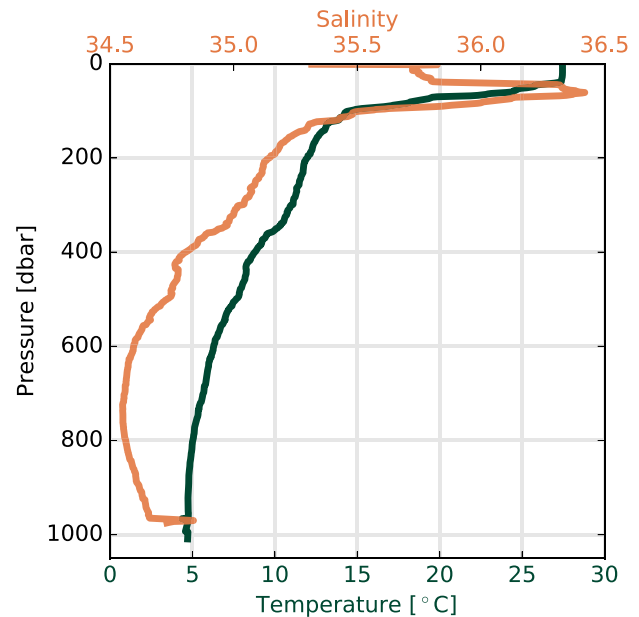


Fig. 1. Vertical profiles of temperature (green) and salinity (orange) at approximately 4°N 38°W on 2008/04/17, from the Brazilian PIRATA hydrographic database. Only the data that passed the quality control procedure recommended by the EuroGOOS are shown. The sharp change of salinity around 950 dbar is due to sensor errors, but it was missed by the EuroGOOS automatic quality control procedure. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2. Data

The different approaches to QC are illustrated in this manuscript applying CoTeDe on the historical hydrographic conductivity–temperature–depth (CTD) dataset from the Prediction and Research Moored Array in the Atlantic (PIRATA, Bourlès et al., 2019). This dataset is composed of 194 CTD profiles, sampled between 1998 and 2011, with over 380,000 measurements of pressure, temperature, and salinity. Fig. 1 illustrates one of these profiles, showing temperature (green) and salinity (orange) as a function of pressure. The CTD profiles used here were collected in the surroundings of the PIRATA buoys in the Western Tropical Atlantic, between 15°N 38°W and 19°S 34°W. Changes in the format of the data were handled using the package pySeabird (Castelão and Irber, 2016), which extracts the data and metadata and combines them into a comprehensive and standard structure.

3. Methods

The package CoTeDe implements all QC tests that will be discussed in this section, providing a convenient framework to compare the flagging performance of different approaches. Traditional QC procedures conforming with the recommendations from GTSP (UNESCO-IOC, 2010), EuroGOOS (DATA-MEQ working group, 2010) and Argo (Wong et al., 2020) are also available and, hence, used as the reference to assess the performance of the proposed Anomaly Detection technique. Note that the goal here is to compare the performance of different automatic QC procedures; thus, only the automatic QC component from these three groups (GTSP, EuroGOOS, and Argo) is considered, neglecting the manual inspection that GTSP, EuroGOOS, and Argo apply to their products.

3.1. Traditional quality control

Oceanographic data have been traditionally quality controlled using a sequence of independent tests, each looking for a known signature, or

characteristics, of spurious measurements. These tests can be grouped into two types. The first type checks for missing or invalid information such as date, time, and geographic coordinates, while the second type compares some characteristics of the data against a previously defined window of acceptable values. Although unfeasible measurements are trivial to detect by the so called “global range” test, the identification of bad measurements within acceptable magnitude is much more challenging and requires more complex tests. One solution to QC bad measurements that are not flagged by the global range test is to project the original data into dimensions that emphasize specific characteristics of bad measurements, with the goal to obtain a new space where good and bad data spread apart from each other in distinct clusters. Each projection is hereinafter referred as a *feature* of that measurement.

In this manuscript I will use $\mathbf{x}$ to describe a set of measurements of a given variable x , such as temperature, and y_n will be the n th feature of $\mathbf{x}$, so that $y_n = y_n(\mathbf{x})$. The index i refers to one specific measurement, such that x_{i-1} (x_{i+1}) is the previous (following) measurement in respect to x_i . Given that, some examples of features widely used are:

Rate of Change: Evaluates the change in respect to the previous measurement as

$$y_{r_i} = x_i - x_{i-1}. \quad (1)$$

Gradient: Evaluates the rate of change in respect to the neighboring points, defined as

$$y_{g_i} = \left| x_i - \frac{(x_{i+1} + x_{i-1})}{2} \right|. \quad (2)$$

Spike: Evaluates how different a measurement is in comparison with its neighbors, defined as

$$y_{s_i} = y_{g_i} - \left| \frac{(x_{i+1} - x_{i-1})}{2} \right|, \quad (3)$$

where y_{g_i} is the *gradient* given by Eq. (2).

Tukey 53H: Evaluates how different a measurement is in comparison to the low-frequency tendency of the data series. This feature takes advantage of the robustness of the median to create a smoother data series that is used as reference (Goring and Nikora, 2002), with the following procedure:

1. $x^{(1)}$ is the median of the five points from x_{i-2} to x_{i+2} ;
2. $x^{(2)}$ is the median of the three points from $x_{i-1}^{(1)}$ to $x_{i+1}^{(1)}$;
3. $x^{(3)}$ is defined by the Hanning smoothing filter

$$\frac{1}{4} \left(x_{i-1}^{(2)} + 2x_i^{(2)} + x_{i+1}^{(2)} \right);$$

4. Finally,

$$y_{t_i} = \frac{|x_i - x_i^{(3)}|}{\sigma}, \quad (4)$$

where σ is the standard deviation of the low-pass filtered data.

Climatology: Evaluates the bias between the observed measurement and a climatology, normalized by the expected variability in that point and time, using the relation

$$y_{c_i} = \frac{|x_i - \langle x_i \rangle|}{\sigma_i}, \quad (5)$$

where $\langle x_i \rangle$ is the climatology, and σ_i is the standard deviation of the observations used to create the climatology at the position i .

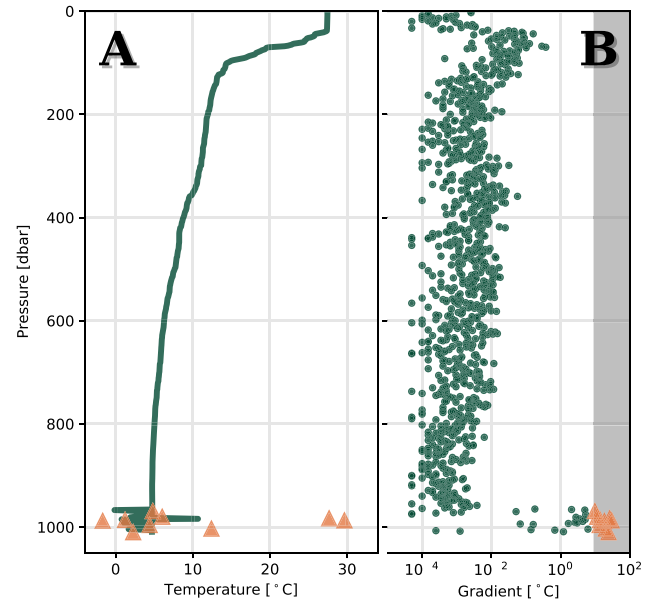


Fig. 2. (A) Temperature profile of station #10 from the PIRATA-X cruise; the green line is the data approved by the gradient test, while the orange triangles failed. (B) The same data plotted in respect to the *gradient* (Eq.: (2)) show a distinct scale for the bad values. The gray area delimits the threshold according to GTSP.

Table 1

Quality control flags recommended by IOC–UNESCO, and adopted in CoTeDe.

Flag	Meaning
0	No QC was performed
1	Good data
2	Probably good data
3	Probably bad data
4	Bad data
9	Missing data

Commonly used climatologies are the World Ocean Atlas (WOA, Locarnini et al., 2010; Antonov et al., 2010; Boyer et al., 2019) and the CSIRO Atlas of Regional Seas (CARS, Ridgway et al., 2002)

Fig. 2 illustrates the traditional QC approach using the gradient test, which projects the data into the *gradient* space and applies a threshold. Measurements classified as bad using the threshold specified by GTSP are marked in orange triangles. Spurious measurements that appear clustered together in the temperature profile (panel A) spread apart as a distinct cluster in *gradient* space (panel B), which allows most of the bad measurements to be captured by the threshold (panel B, gray shading). However, closer inspection of Fig. 2 suggests that the gradient threshold missed some measurements near 1000 dbar that should have been classified as bad.

After applying the traditional QC approach using the tests listed above, each measurement receives a quality flag following the recommendations of the Intergovernmental Oceanographic Commission of UNESCO (Table 1), a widely adopted flag standard (UNESCO-IOC, 2010; DATA-MEQ working group, 2010; SeaDataNet, 2010; Wong et al., 2020). The final flag of each measurement is the maximum flag value obtained among all performed tests, i.e., a given measurement is only considered good (flag 1) if it passed (it was flagged as 1) in all tests. Hereinafter, I will use the term “false-positive” to refer to data that was flagged as good data, but it should have been flagged as bad. In contrast, “false-negative” will refer to data that was flagged as bad data, but it should have been flagged as good.

3.2. Anomaly detection

Anomaly Detection is a classification technique used to discriminate commonly observed data from anomalies. While other classification methods try to describe each one of the classes under consideration, the Anomaly Detection approach focuses on recognizing the common behavior of the data. By assuming that spurious measurements are anomalies, the Anomaly Detection provides a solution to quality control that is less sensitive to the sample size of bad data. Further, because the Anomaly Detection does not rely on specific patterns to recognize bad data, this technique has the potential to promptly identify unprecedented anomalous patterns, while other techniques would require to explicitly learn the new pattern first.

The Anomaly Detection approach is not new to environmental sciences. Bettencourt et al. (2007) applied this technique to a network of inland synchronous sensors, identifying bad samples even when the spurious measurements had feasible magnitudes. A method that evaluates the data by the magnitude itself requires a stationary statistics or a sampling rate sufficiently high to overcome the environment changing trend (Bettencourt et al., 2007). To bypass the non-stationarity issue, Bettencourt et al. (2007) looked at the distribution of the differences between measurements from neighboring sensors, as opposed to the measurement itself, and detected the anomalies using a p-test.

An alternative approach was proposed by Hill and Minsker (2010), who used sequential measurements from a single sensor to compute auto-regressive models, including a perceptron type of artificial neural network, to predict the following value in the timeseries. The authors used a pre-defined confidence interval on the prediction to obtain the range of tolerance, so a value outside that range would be an outlier, and assumed to be a bad measurement. This solution can handle small gaps in the data as long as the sampling rate is sufficiently high, but it also requires either stationarity or frequently updating the model parameters according to the environmental conditions.

The solutions presented by both Bettencourt et al. (2007) and Hill and Minsker (2010) could be potentially applied to measurements from ocean moorings with high sampling rate. However, their approaches may prove challenging to implement for most hydrographic profiles for two main reasons. First, duplicate measurements in the open ocean are rare and mostly restricted to modern CTD casts. Second, it would be impractical to update the model parameters often enough to account for the vertical variability within an individual profile. The alternative that I propose here maps the original variables into spaces that emphasize different characteristics of the measurement, and then evaluates how anomalous the mapped measurements are, instead of evaluating the measurement itself.

In the context of machine learning, the features of a state are a set of attributes (or characteristics) relevant to predict a solution (or the state value). Features are a building block of classification problems since they aggregate information otherwise not available from the raw state description (Russell and Norvig, 2016). The tests used in traditional QC methods (Section 3.1) are the most natural choice of features for the Anomaly Detection approach since they were designed to emphasize known characteristics of bad data. Here, I use the set of features defined in Section 3.1, but instead of restricting the QC to a binary classification (good or bad) based on a hard threshold for each individual feature, the feature values are combined into a continuous scale. For example, in the traditional approach a *gradient* of 9.99 °C would result in the measurement being simply classified as “good” according to the GTSP threshold (≤ 10 °C). However, this classification would not provide any context for how confident we should be in the classification (see Fig. 2B). In contrast, the Anomaly Detection would consider how uncommon it would be for a “good” measurement to have a *gradient* of 9.99 °C and use that in combination with other feature values to provide a more flexible and multidimensional classification criterion. The full Anomaly Detection procedure implemented in CoTeDe is explained in detail as follows.

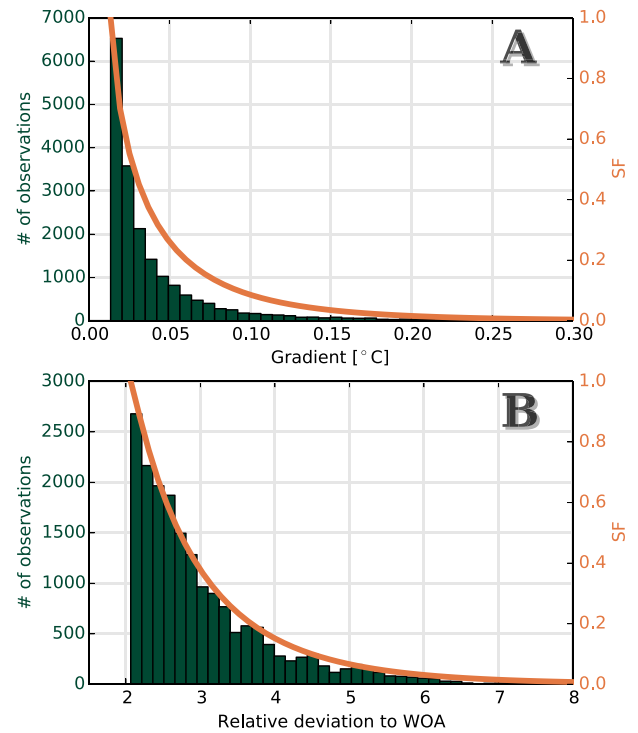


Fig. 3. (A) Distribution of the top 10% *gradient* test results (green), and the respective survival function (orange). (B) Distribution of the top 10% *climatology* comparison test (green), and the respective survival function (orange). Only the data approved by the EuroGOOS QC procedure is considered. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The first task is to characterize the typical behavior of the data by estimating a probability density function (PDF) for each feature (y_n). Since the goal is to identify anomalies, and bad measurements are usually much less frequent than 1%, any feature value below the 90th percentile is considered common, i.e., there is no evidence that the measurement is bad based on that individual feature. Then, the PDF of each feature is estimated using only the top 10% values of y_n , allowing for a better fit in the range of interest.

After testing multiple analytical functions, the exponentiated Weibull distribution resulted in the best overall fit and it was used for all features. The exponentiated Weibull function is defined as:

$$\text{PDF}(y|\lambda, \alpha) = \frac{k}{\lambda} \left(\frac{y}{\lambda}\right)^{k-1} \left[1 - e^{-\left(\frac{y}{\lambda}\right)^k}\right]^{\alpha-1} e^{-\left(\frac{y}{\lambda}\right)^k}, \quad (6)$$

where k , λ , and α are the fit parameters, and y is a feature of the variable to be classified, for example the *gradient* (Eq. (2)) of temperature measurements $y_g = y_g(\mathbf{T})$.

The respective survival function (SF) of the estimated PDF is used to quantify how anomalous a certain measurement is. For a feature y_n , $\text{SF}_n(y_{n_i})$ gives how frequent a valid measurement x was observed with $y_n \geq y_{n_i}$. Thus, the higher y_{n_i} , the smaller $\text{SF}_n(y_{n_i})$, and more anomalous is x_i in the perspective of the feature y_n . Fig. 3 illustrates the distribution of the 10% highest values of *gradient* (A) and *climatology* (B) of temperature from the entire PIRATA dataset along with the respective survival functions. Panel A shows that only 10% of the observations had a *gradient* higher than 0.013 °C. Therefore, values equal or below 0.013 °C are perceived by the model as common data, lacking any indication of being an anomaly. Now, let's consider the case where $y_g = 0.1$. Fig. 3A shows that $\text{SF}_g(y_g = 0.1) = 0.077$, hence there is a 7.7% chance of obtaining a sample x with $y_g \geq 0.1$.

Assuming that the features are independent, the probability of observing a good measurement (x_i), characterized by a set of feature

values $\{y_g(x_i), y_c(x_i), y_s(x_i), \dots\}$, or a more rare scenario, is given by the product of the individual probabilities

$$p_i = \prod_n \text{SF}_n(y_{n_i}), \quad (7)$$

where y_{n_i} is the n th feature of the measurement x_i .

Lastly, it is necessary to define a probability threshold ($\bar{p}$) in order to distinguish good data from anomalies. To obtain this threshold, a first guess was calculated by applying the traditional QC approach (Section 3.1) to the PIRATA dataset following the automatic QC procedure recommended by EuroGOOS. The data flagged as good following the EuroGOOS procedure were then randomly split into three subgroups: fit, test, and error estimate, with 60%, 20%, and 20% of the valid observations, respectively. Additionally, the data flagged as bad by the EuroGOOS procedure were randomly split in half, and each half was included in the test and error estimate groups. The PDF coefficients were adjusted based only on the fit group, hence expected to be mostly, if not fully, composed of good data. As a consequence, the survival functions are an indicative of how common a given value is among good measurements. Using the survival functions, the threshold $\bar{p}$ was then calculated to minimize the sum of false positive and false negative cases considered in the test group. Finally, the error of the Anomaly Detection approach was estimated by applying $\bar{p}$ on the error estimate group, and since the data on the error estimate group is not used in the adjusting procedures, this is an unbiased error estimate.

In summary, the Anomaly Detection will flag a measurement x_i as good (1) if the probability p_i is greater than the threshold $\bar{p}$. Otherwise, x_i is flagged as probably bad (3).

4. Results and discussion

The traditional QC approach consists of a sequence of independent tests that compare the values of individual features against fixed thresholds. Because each test is applied independently, the traditional QC has limited degrees of freedom, and its calibration is restricted to either increasing or decreasing the thresholds. Thus, the traditional approach has an intrinsic conflict when choosing thresholds, since a high (low) threshold favors false positives (negatives). Fig. 2 illustrates this limitation, where the GTSP threshold for the gradient test (10 °C) misses spurious measurements near 1000 dbar in order to accommodate for good measurements that have gradients of similar magnitude in the thermocline region (upper 200 dbar).

Since failing in one test is sufficient to flag the data as bad, the strategy usually adopted by data centers is to calibrate the traditional QC approach such that it minimizes false negatives. Therefore, when the traditional QC flags a measurement as bad, the classification is generally correct, but that comes with the price of also having bad measurements being flagged as good (false positives). In this context, improvements in QC methods aiming to reduce the burden on human expert QC should target false positives.

Identifying bad measurements that have unfeasible values is a trivial procedure, thus all results and considerations presented below are for the PIRATA dataset after discarding the points that failed on the global range test of the traditional QC approach (Section 3.1), which removes 0.13% of the measurements from the full dataset. Fig. 4 projects temperature measurements into *climatology* and *Tukey 53H* feature space. The observations are flagged as good (green) or bad (orange) after applying the traditional QC approach following the EuroGOOS recommendations for automatic real-time QC (see DATA-MEQ working group, 2010). Since the real-time EuroGOOS flagging does not include Tukey 53H and climatology tests, the classification is independent of the projection shown in Fig. 4.

The two gray boxes illustrate the region above the chosen thresholds for the Tukey 53H and climatology tests (1.5 and 6, respectively). The Tukey 53H test (panel A) agrees with the EuroGOOS real-time automatic classification, capturing only bad data (orange). Thus, it

would be redundant to aggregate the Tukey 53H test with this particular threshold to the set of tests used by the real-time EuroGOOS. In contrast, the climatology test (panel B) would flag as bad some measurements that the real-time EuroGOOS had classified as good data (note that the gray box in panel B overlaps with some of the histogram in green).

Considering only the data flagged as good by the real-time EuroGOOS tests (green), a climatology test with a threshold of 6 would flag an additional 0.06% of the data as bad, while a threshold of 3 (as recommended by GTSP), would increase that to 1.05%, which is one order of magnitude higher than the expected for a normally distributed dataset. Manual classification of the PIRATA dataset confirms that a climatology threshold of 3 (6) results in more than 1% (0.05%) of false negatives in the dataset, i.e., the climatology test recurrently mistakes uncommon events for bad data. Most of these failures in the automatic classification using the climatology test were due to warm anomalies, which is corroborated by the median of the feature *climatology* which is equal to 0.3 °C. In other words, 50% of the PIRATA measurements were at least 0.3 °C warmer than the climatological mean.

The climatology test in the traditional QC approach assumes a normally distributed dataset with stationary statistics. If this is violated, such test would systematically reject good data, potentially introducing biases in the final product. Another potential issue is that in regions with insufficient historical observations the climatological variability might not be properly captured. One way to address this problem would be to account for the uncertainty in the climatological mean. Regions with fewer historical samples would likely have a larger standard error of the mean, corresponding to higher uncertainty in the climatology. Thus, the threshold for the climatology test should take that into account by subtracting the standard error of the mean from the climatology difference.

While it is challenging to define an optimal threshold for each unidimensional projection when good and bad data overlap (4A–B), the projection into a bidimensional space shows a clear polarization between the two classes (4C). Visual inspection of Fig. 4 suggests that the black dashed line would be a better criterion to separate the two classes (good and bad), but such slope is not possible when evaluating the features individually. The traditional QC procedure is equivalent to expanding or shrinking the gray rectangles, but always keeping the same shape. For example, the upper edge of the good data cluster (panel C, green) would be flagged as bad by the traditional approach with a threshold of 6 for the climatology, but manual classification suggests that these are valid measurements resulting from anomalous natural events. Additionally, the sparse bad data (panel C, orange dots) in the middle of the cluster of good data (panel C, green cloud), that have low values of *climatology* and *Tukey 53H*, illustrates how the projections can be orthogonal, thus reinforcing the need for multiple features to identify spurious data. A multidimensional space analysis allows for a criterion with more degrees of freedom and, with a careful set of features, the good data is identified as a distinct cluster. A major advantage of QC by human experts comes from context awareness (Smith et al., 2012). By taking more information into account, the expert can evaluate cases that are not obvious to classify at a first glance. Techniques such the Anomaly Detection, fuzzy logic (Timms et al., 2011; Morello et al., 2011, 2014), and Bayesian networks (Smith et al., 2012) can better mimic the human expert by combining features into a multidimensional criterion, which leads to higher context awareness and allows for more intricate classification in comparison to multiple unidimensional tests.

Fig. 5 illustrates a profile of temperature approved by the automatic real-time EuroGOOS procedure. The zoom around 724 dbar shows a questionable abrupt change on the profile. The feature values are not large enough to be individually considered bad data by the traditional QC thresholds (see Table 2), but the Anomaly Detection approach identifies this measurement as a distinct structure in the profile (Fig. 5, orange). While traditional QC does a good job avoiding false negatives,

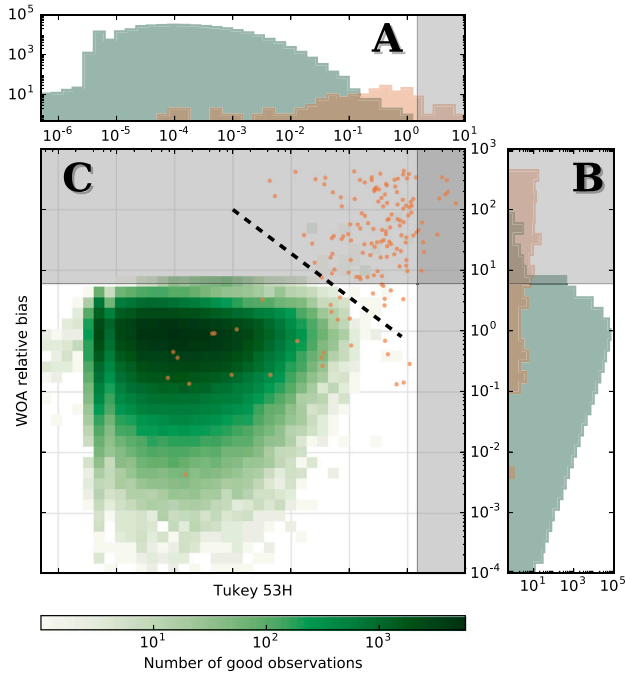


Fig. 4. Observations of the PIRATA-Brazil hydrography in respect to *Tukey 53H* (A) and *climatology* (B). The good data are in green, and the bad data in orange, according to the EuroGOOS recommendation for real-time. The gray boxes delimit *Tukey 53H* and *climatology* above 1.5 and 6, respectively. The black dashed line is an approximate threshold between the good and bad data clusters. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

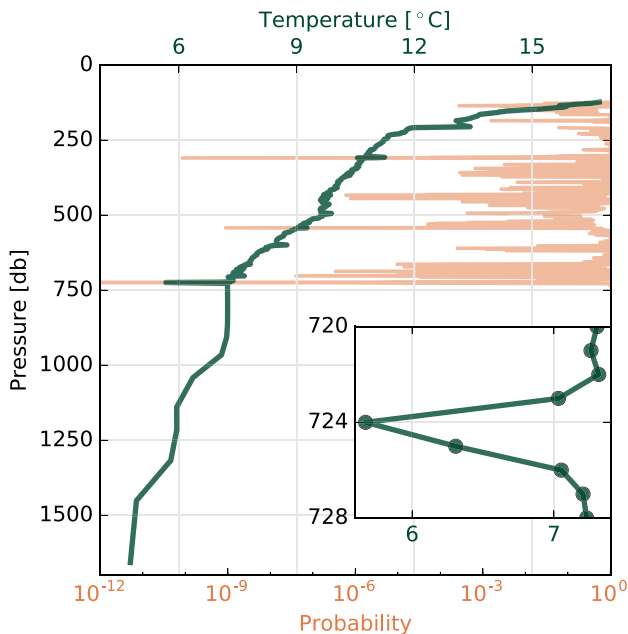


Fig. 5. Temperature measurements from the cruise PIRATA-X, profile 10. The data approved by the EuroGOOS procedure is shown in green, and the probability of being a good data, according to the anomaly detection, is shown in orange. The small panel shows a zoom in the temperature between 720 and 728 dbar. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2

Observed temperatures and respective quality control test results for the samples 600 to 602 of the profile 10 of cruise PIRATA-X. This interval is also shown in the zoom of Fig. 5. The last column shows the thresholds suggested by the EuroGOOS procedures.

	x_{600}	x_{601}	x_{602}	thr.
Pressure [dbar]	723	724	725	
Temp. [°C]	7.03	5.67	6.31	
Gradient	0.54	1.00	0.05	3°
Spike	-0.28	0.64	-0.64	2°
Climatology	0.75	3.11	1.28	6
Tukey 53H	0.01	0.28	0.15	1.5
Anom. det.	2e-6	1e-20	9e-7	

the Anomaly Detection technique complements that by identifying false positives.

The Anomaly Detection can be calibrated to reproduce the automatic EuroGOOS flagging (including the climatology test). This calibration resulted in a classification error of approximately 0.4%; however, since most of the errors are due to a known false positives bias in the traditional QC techniques, 0.4% is an overestimate of the Anomaly Detection errors. To properly evaluate the performance of each method, a better reference flagging was obtained through active learning as described below.

In the first iteration, the Anomaly Detection was calibrated and evaluated assuming the EuroGOOS as the truth. The severity score of each supposedly misclassified measurement is quantified by the difference between the probability threshold ($\bar{p}$) and the estimated probability of being a good data (p_i). The rationale behind it being that in order to avoid a misclassification with large $|p_i - \bar{p}|$, one would require greater changes in the classification parameters that ultimately defined p_i , hence that mistake would be in greater disagreement with the criteria used than one with a small $|p_i - \bar{p}|$. The severity score drives an iterative process: the worst mistake is manually evaluated first, and the flag is confirmed or corrected, so the reference is updated, the anomaly detection recalibrated, and the severity score redefined. The human effort is hence optimized in such way that most critical errors are classified first while the calibration converges. The performance of each recalibration is evaluated using an independent dataset, the error subset described in Section 3.2, and the iteration process ceases once the error in the error subset stabilizes or increases, hence avoiding overfitting. Such active learning provides a better reference classification without the need of manually processing the whole dataset, and could leverage existing tools for manual QC (e.g., Schlitzer, 2002; Thadathil et al., 2001) by making the human expert effort more efficient.

Table 3 shows the performance of the different QC procedures available in CoTeDe. The GTSP and EuroGOOS for real-time (without climatology) achieved the lowest rate of false negatives, but also had the highest rate of false positives. Including the climatology on those procedures reduces the ratio of false positives while increasing the rate of false negatives. For the GTSP that was critical, resulting in the worst overall performance, misclassifying 1% of the dataset. The fuzzy logic approach of Morello et al. (2014) further reduces the false positives for the cost of increasing the overall classification with a tendency to flag good data as bad. The Anomaly Detection achieved the best overall performance with 2 classification errors per 10,000 measurements, and reducing by half the errors by EuroGOOS for real-time — which is currently accepted as the best procedure. It is worth noting that the error by the Anomaly Detection was estimated from the error subset, hence independent of the calibration.

Gronell and Wijffels (2008) also explored the concept of identifying bad data by searching for outliers through multiple features, employing an equivalent of the climatology test on each feature. That was a major improvement over the traditional QC since the local natural variability scaled the test threshold, avoiding to use one constant for the whole ocean. Despite the differences in the methodologies, there is some

Table 3

Ratio of errors per 10,000 measurements of different QC approaches estimated on the PIRATA hydrography dataset approved on the global range test, i.e. after removing the unfeasible values. For reference, manual QC resulted the ratio of 8.8 bad data per 10,000 measurements.

QC procedure	False bad	False good	Total error
GTSP (w/clim.)	97.4	2.9	100.3
GTSP real-time	0.0	5.2	5.2
EuroGOOS (w/clim.)	4.3	2.8	7.1
EuroGOOS real-time	0.0	4.1	4.1
Morello2014	11.6	2.7	14.3
Anomaly detection	0.2	1.8	2.0

conceptual equivalence between that technique and what I propose here. The main improvement from Anomaly Detection comes from the multidimensional criterion, but it also avoids the normal distribution assumption and drastically reduces the required manual QC effort.

The QC methodology introduced here allows the inclusion of new features, which may provide additional perspectives on the behavior of data and may help to identify measurement errors. For example, some specific cruises analyzed here persistently lacked data in the first tens of meters, where the CTD was lowered faster than recommended. A human expert would note the improper operating procedures and would be more likely to flag data as bad even with the smallest indication of it being bad. The Anomaly Detection could mimic that analysis by aggregating two new features: shallowest measurement in a profile, and descending rate, for example. Morello et al. (2014) uses for other purposes the time since the last calibration, while Gronell and Wijffels (2008) introduce several other features, which could all be added in this example. In the case that any of those characteristics are too anomalous, it would contribute for the total uncertainty probability (p in Eq. (7)), hence a smaller *gradient* or *spike* would be sufficient to exceed the acceptable $\bar{p}$. The Anomaly Detection as proposed here is a quantitative way to accumulate and combine different aspects of the data for a non-linear classification, allowing for decisions with deeper context awareness.

An intrinsic byproduct of the Anomaly Detection is to define how uncommon a given scenario is. Yao et al. (2010) discuss the importance of identifying real-time anomalous, but valid, measurements for management response, such as to detect an algae bloom. This concept raises new possibilities for autonomous sampling systems (e.g. Argo floats, Spray underwater gliders, and Saildrones). An intelligent sensor running an onboard real-time quality control could be setup to increase the sampling rate once a threshold on the probability of occurrence is reached. That would minimize the losses from bad samples, as well as increase the sampling resolution of uncommon, but physical, events. This would be a major improvement on the optimization effort of the observing systems. For example, a Spray underwater glider could stop in a place for an extra dive, before keeping its pre-planned mission, once it detected something rare. An Argo float could anticipate its cycle and redo a profile if the previous measurements were unexpected. A Saildrone could adjust its course and increase the sampling rate once crossing a sub-mesoscale front. It is common to keep subsurface moorings over a year at sea without any communication, so any interesting event would only be found after recovering the equipment. An intelligent adjusting sampling rate would increase the spectrum coverage of autonomous sensors, with the same storage memory and power budget.

5. Concluding remarks

Machine learning techniques provide fascinating strategies to automatically classify data by employing supervised learning, which is based on the principle of training the classification system with some known answers. Such calibration usually requires sufficient data to

statistically represent each class, but the better the measuring procedure, the smaller the amount of bad data. The cluster undersampling technique can overcome the contrast in relative sampling sizes (Rahman et al., 2014), but having too few bad samples would still be a serious limitation for the usual approach of identifying each class, i.e., recognizing a bad measurement in the same way that it recognizes a good measurement. In addition, observations in the open-ocean are typically sparse in space and time, and overlapping measurements are rare. Thus, most machine learning techniques, such as neural networks, although powerful, are not an adequate approach to QC open-water oceanographic data due to intrinsic characteristics of oceanographic measurements.

The novel approach based on the Anomaly Detection technique that I proposed here strongly impacts the QC of oceanographic data in twofold. First, it optimizes the expert effort by driving the manual evaluation towards the most dubious measurements first, allowing the experts to efficiently handle the ever increasing amount of measurements in the oceans. This exact principle is currently being employed by the International Quality controlled Ocean Database¹ (IQuOD) to optimize their expert QC effort. Second, it combines multiple characteristics of each measurement for a deeper decision making, resulting in a higher context awareness for more intricate classification. The same Anomaly Detection is not limited to QC, but could also be used to guide self adjusting sampling platforms, increasing the spectrum coverage of the measurements.

The Python package CoTeDe (Castelão, 2020) is an open source framework that enables easy application of the current state of the art QC techniques on oceanographic measurements, with the possibility to customize the set of tests to be used.

Computer code availability

The Anomaly Detection technique introduced here was implemented in the open-source QC framework CoTeDe under a 3-Clause BSD license, and it has been publicly available since 2013 as a git repository at <http://cotede.castelao.net>.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.cageo.2021.104803>.

¹ <http://www.iquod.org>.

² <https://www.mar.mil.br/dhn/chm/oceanografia/bndo.html>.

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